Hybrid Ant Colony Optimization Method for the Traveling Salesman Problem

Udom Janjarassuk
Department of Industrial Engineering, School of Engineering
King Mongkut's Institute of Technology Ladkrabang
Bangkok, Thailand
udom.ja@kmitl.ac.th

2024 9th International Conference on Business and Industrial Research (ICBIR) | 979-8-3503-8302-7/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICBIR61386.2024.10875933

***Abstract*—The traveling salesman problem (TSP) is a classic problem in computer science and operations research which involves finding the shortest possible route that visits a given set of cities. In this paper, we propose a hybrid algorithm for solving such problem. The algorithm combines the ant colony optimization (ACO) method with the 2-opt heuristic to improve the efficiency for solving the TSP. Instances from the TSPLIB were used to test the algorithm. The results showed that the hybrid ACO algorithm was more effective in solving the TSP as compared to the traditional ACO or the 2-opt heuristic methods.**

***Index Terms*—traveling salesman problem, ant colony optimization, hybrid method, 2-opt heuristic, optimization, algorithm**

I. Introduction

The traveling salesman problem (TSP) is an example of a combinatorial optimization problem, which involves finding the optimal solution from a finite set of possible solutions. The TSP is often used as a model for real-world problems such as routing delivery trucks or planning the most efficient sales routes for a salesperson [1]. The TSP is difficult to solve particularly for large sets of cities.

There are a number of algorithms that have been developed to solve the TSP [1], [2], including exact algorithms, approximation algorithms, and heuristics. For examples, dynamic programming, branch and bound algorithm, nearest neighbor search, genetic algorithm (GA), and ant colony optimization (ACO) [3]–[5].

Due to the complexity of the TSP, heuristic algorithms have been widely used to solve large practical problems. There are many well known heuristics, including methods such as greedy algorithms, local search, and genetic algorithm. Ant colony optimization is one of the most applicable algorithms for solving many combinatorial optimization problems including the TSP. It was first proposed by A. Colorni, M. Dorigo, and V. Maniezzo in 1991 [6] known as the Ant System (AS). Since then, many variations of ACO algorithms have been developed such as Elitist Ant System (EAS), MAX-MIN Ant System (MMAS), and Ant Colony System (ACS) [7], [8].

Hybrid ACO algorithms combine elements of ACO with other optimization techniques, such as evolutionary algorithms, local search, or artificial neural networks, in order to improve the performance and scalability of ACO. The efficiency of hybrid ACO algorithms for solving the traveling salesman problem can vary depending on the specific design of the algorithm and the characteristics of the problem instance. In general, hybrid ACO algorithms tend to be more efficient and scalable than pure ACO algorithms, as they can take advantage of the strengths of multiple optimization techniques and can adapt to different problem characteristics.

In this paper, we propose a hybrid ACO algorithm which combines the ACO with the 2-opt heuristic to improve the performance and scalability of ACO for solving the TSP. The goal of this study is to focus on the efficiency of the hybrid algorithm as compared to the traditional ACO algorithm. The remaining of the paper is organized as follows: Section II provides an overview of the 2-opt heuristic and the ACO methods, and discusses the proposed hybrid ACO algorithm. Computational results are reported in Section III, followed by a conclusion in Section IV.

II. Methodology

A. 2-opt Heuristic

The 2-opt heuristic is a local search algorithm for solving the traveling salesman problem. It was first proposed by Croes in 1958 [9]. The algorithm removes two arcs from a route and reconnects the route in different order. If the modification has led to a shorter total distance, then the current route is updated. The algorithm repeats until no more improvements are found or until a pre-specified number of iterations is met. The idea of the algorithm is illustrated in Fig. 1.

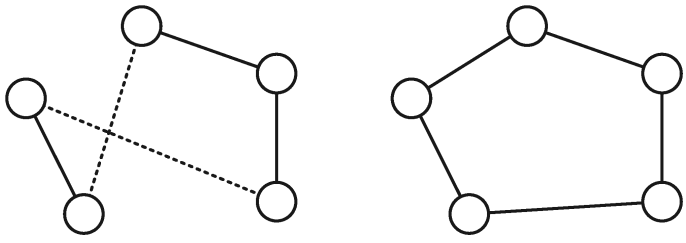


Fig. 1. 2-opt Heuristic.

B. Ant Colony Optimization

The ant colony optimization algorithm mimics the behavior of ants in finding food for constructing solution of an optimization problem. When a food source is discovered,

Authorized licensed use limited to: BMS College of Engineering. Downloaded on November 05,2026 at 10:21:58 UTC from IEEE Xplore. Restrictions apply.

pheromone will be released by ant along the trail to the nest. Pheromone is a chemical substance which will evaporate over time. As ants travel along the trail, pheromone will be intensified by ants. When multiple trails are available, ants will more likely to follow the trail with a stronger pheromone intensity. Pheromone on a shorter trail will be intensified more often by ants, resulting for the longer trails to be abandoned.

Equation (1) provides the formula to compute the probability of ant traveling from city i to city j, where j is in the set of selectable cities S. τ_{ij} is the pheromone value of the path from city i to city j, and η_{ij} is the visibility of ant to travel from city i to city j. The visibility is defined as $\eta_{ij} := 1/d_{ij}$, where d_{ij} is the distance from city i to city j. The parameters α and β are the influence levels on pheromone value and visibility respectively.

$$p_{ij} := \frac{\tau_{ij}^{\alpha} \cdot \eta_{ij}^{\beta}}{\sum_{k \in S} \tau_{ik}^{\alpha} \cdot \eta_{ik}^{\beta}} \quad \forall j \in S, \tag{1}$$

Pheromone values are updated in each iteration by evaporation and intensification. The evaporation process reduces the value of pheromone by an evaporation rate ρ according to (2), and the intensification process increases the pheromone value by an amount of Δ according to (3). Note that the intensification process is only updated for the pheromone values associated with the best solution x^* in each iteration. The parameter n in (2) and (3) is the number of cities.

$$\tau_{ij} := (1-\rho)\tau_{ij}, \qquad \forall i, j \in \{1, ..., n\}, \tag{2}$$

$$\tau_{ix^*(i)} := \tau_{ix^*(i)} + \Delta, \qquad \forall i \in \{1, ..., n\}, \tag{3}$$

A scheme of the ACO algorithm is given in Algorithm 1. The pheromone values are initialized at the beginning of the algorithm. In each iteration, m artificial ants are created to construct m routes for the TSP. The construction of route starts from city 1 and selects the next city to visit according to the probability given in (1) until the set of selectable cities S is empty. Then the pheromone values are updated according to the evaporation process and intensified for the values associated with the best solution. The algorithm stops when the stopping criterion is satisfied.

C. Hybrid ACO

The proposed hybrid ACO is a combination of the 2-opt heuristic and the traditional ACO as described in sections II-A and II-B. As shown in Algorithm 2, the parameter γ is used to control the probability of choosing the 2-opt heuristic method for the route construction for an ant. If a randomly generated number from a uniform distribution on the interval $[0,)$ is less than γ, then the 2-opt heuristic method is applied for the route construction instead of the ACO method. Pheromone values are updated by the evaporation and intensification processes in each iteration. The algorithm repeats until the stopping criterion is met.

Algorithm 1 Ant Colony Optimization for TSP

```
Initialize pheromone values
repeat
  for ant k ∈ {1, ..., m} do
    i := 1
    S := {2, ..., n}
    repeat
      Select city j ∈ S with probability p_ij
      S := S − {j}
      i := j
    until S = ∅
  end for
  for all i, j do
    τ_ij := (1 − ρ)τ_ij
  end for
  for all i, j in the best route do
    τ_ij := τ_ij + Δ
  end for
until stopping criterion is satisfied
```

Algorithm 2 Hybrid ACO for TSP

```
Initialize pheromone values
repeat
  for ant k ∈ {1, ..., m} do
    if random[0,1) < γ then
      Construct a 2-opt route
    else
      Construct an ACO route
    end if
  end for
  for all i, j do
    τ_ij := (1 − ρ)τ_ij
  end for
  for all i, j in the best route do
    τ_ij := τ_ij + Δ
  end for
until stopping criterion is satisfied
```

III. Computational Results

The proposed hybrid algorithm is tested by using 15 instances from the TSPLIB [10]. The name of the instances and the number of cities are listed in the first two columns in Table I respectively. The optimal values of the instances are given in column three. The smallest instance has 14 cities and the largest instance has 1304 cities. The values of parameters used in the experiments are as following: the number of ants $m = 50$, pheromone influence level $\alpha = 3.0$, visibility influence level $\beta = 5.0$, evaporation rate $\rho = 0.05$, pheromone increment $\Delta = 0.1$, the hybrid probability $\gamma = 0.05$, and the maximum number of iterations is limited to 100. These values are obtained from trial and error which provide a relatively good performance among all tested instances.

Authorized licensed use limited to: BMS College of Engineering. Downloaded on November 05,2026 at 10:21:58 UTC from IEEE Xplore. Restrictions apply.

TABLE I
RESULTS OF THE OBJECTIVE VALUES AND AVERAGE TIME

Instance	Number of Cities	Optimal	Best Value	Best Gap(%)	Standard Deviation(%)	Average Time(sec)
a280	280	2579	2697	4.58	0.63	100.51
att48	48	10628	10808	1.69	0.34	3.28
berlin52	52	7542	7686	1.91	0.79	3.86
burma14	14	3323	3323	0.00	0.00	0.27
ch130	130	6110	6342	3.80	0.69	23.33
d198	198	15780	16263	3.06	0.00	52.98
eil51	51	426	438	2.82	0.07	2.86
fl417	417	11861	12457	5.02	0.01	225.63
gil262	262	2378	2567	7.95	0.50	90.71
gr202	202	40160	42690	6.30	0.18	55.17
lin318	318	42029	45557	8.39	0.84	134.23
p654	654	34643	37072	7.01	0.93	550.01
rl1304	1304	252948	285865	13.01	0.80	2257.62
ulysses16	16	6859	6859	0.00	0.05	0.38
ulysses22	22	7013	7013	0.00	0.04	0.69

The computer used for the experiments was an Intel Core i5 processor running at clock speed 2.9GHz. Each instance was solved for 10 replications and the results were provided in Table I. From the table, the best objective value from 10 replications for each instance is listed in column four, and its optimality gap (in column five) was computed (in percentage) from the known optimal value (from column three). From the results, the hybrid ACO algorithm was able to solve the three smallest instances till optimal. The optimality gaps for most instances were within 5%, and the gap for the largest instance was 13.01%. Column six showed the standard deviation of the objective values from 10 replications in percentage. The deviations were less than 1% in all tested instances, which implied consistency of the algorithm. The last column in the table was the average solution time (in seconds) for each instance.

To compare the performance of the hybrid ACO algorithm with the 2-opt heuristic and the traditional ACO algorithm, additional experiments were performed for both methods by using the same settings. The average optimality gaps from each method were reported in Table II. From the results, the hybrid ACO algorithm outperformed the 2-opt heuristic and the traditional ACO algorithm in almost all cases.

TABLE II
RESULTS OF THE AVERAGE GAPS

Instance	Average Gap (%)		
	Hybrid	2-opt	ACO
a280	5.05	12.04	7.23
att48	2.23	5.82	2.99
berlin52	3.25	7.38	5.14
burma14	0.00	0.00	0.00
ch130	4.97	8.16	4.02
d198	3.06	4.81	9.23
eil51	3.03	7.10	5.70
fl417	5.06	8.72	5.13
gil262	9.05	11.39	8.31
gr202	6.44	8.09	7.79
lin318	8.82	9.91	10.79
p654	8.02	8.58	7.43
rl1304	13.84	12.91	13.61
ulysses16	0.04	0.23	0.07
ulysses22	0.06	0.09	0.09

IV. CONCLUSION

In this paper, we proposed an effective hybrid ACO algorithm to solve the TSP. The algorithm combines the 2-opt heuristic and the ACO methods to achieve better performance compared to the results from each individual method. The hybrid ACO algorithm is capable of solving the TSP effectively for small and medium size problems. However, the efficiency of the hybrid ACO algorithm also depends on other factors such as the specific combination of optimization techniques used, the balance between exploration and exploitation, the choice of parameters, etc. As such, there is still room for improvement for the algorithm in terms of solution quality and computation time, particularly for solving large TSP or difficult instances.

REFERENCES

[1] D. Davendra, *Traveling salesman problem: Theory and applications*. BoD–Books on Demand, 2010.
[2] D. L. Applegate, R. E. Bixby, V. Chvatál, and W. J. Cook, *The Traveling Salesman Problem: A Computational Study*. Princeton University Press, 2006. [Online]. Available: http://www.jstor.org/stable/j.ctt7s8xg
[3] J. Ouyang and G.-R. Yan, "A multi-group ant colony system algorithm for tsp," *Proceedings of the Third International Conference on Machine Learning and Cybernetics*, vol. 1, pp. 117–121, 2004.
[4] J. Yang, X. Shi, M. Marchese, and Y. Liang, "An ant colony optimization method for generalized tsp problem," *Progress in Natural Science*, vol. 18, pp. 1417–1422, 2008.
[5] Y. Wang and Z. Han, "Ant colony optimization for traveling salesman problem based on parameters optimization," *Applied Soft Computing*, vol. 107, p. 107439, 2021.
[6] A. Colorni, M. Dorigo, and V. Maniezzo, "Distributed optimization by ant colonies," *Proceedings of the First European Conference on Artificial Life*, pp. 134–142, 1991.
[7] M. Dorigo and C. Blum, "Ant colony optimization: A survey," *Theoretical Computer Science*, vol. 334, pp. 243–278, 2005.
[8] S. Kefi, N. Rokbani, and A. M. Alimi, "Solving the traveling salesman problem using ant colony metaheuristic, a review," in *Proceedings of the 16th International Conference on Hybrid Intelligent Systems (HIS 2016)*, A. Abraham, A. Haqiq, A. M. Alimi, G. Mezzour, N. Rokbani, and A. K. Muda, Eds. Cham: Springer International Publishing, 2017, pp. 421–430.
[9] G. Croes, "A method for solving traveling salesman problems," *Operations Research*, vol. 6, pp. 791–812, 1958.

Authorized licensed use limited to: BMS College of Engineering. Downloaded on November 05,2026 at 10:21:58 UTC from IEEE Xplore. Restrictions apply.

[10] G. Reinelt, "TSPLIB — a traveling salesman problem library," *ORSA journal on computing*, vol. 3, no. 4, pp. 376–384, 1991.

Authorized licensed use limited to: BMS College of Engineering. Downloaded on November 05,2026 at 10:21:58 UTC from IEEE Xplore. Restrictions apply.